

MTH 377/577 Convex Optimization Assignment

April 13, 2026

Answer all questions. You may submit the assignment in groups of maximum 4 students. Mention the names and roll numbers of all the members in your answerscript. Provide detailed mathematical arguments and explanations- show all formal steps. Your answerscript should be legible and carefully written since this is a take-home assignment.

1. Consider a car retail outlet that is giving away 3 cars (A,B,C) to winners of a contest. There are three winners (1st, 2nd, 3rd) and each gets only one car. Suppose that car A is the most valuable car for all three winners. Car B,C are the second highest valued cars for winners 2 and 3 respectively. Winner 1 values cars B and C equally. How can the cars be assigned efficiently to the three winners? Provide the formal model and solve for the efficient assignment. Write down all the assumptions and show all the steps. (7)
2. Suppose that the following 8 values are observed for a two-dimensional variable: $(3, 2), (4, 0), (-4, 3), (-1, 0), (2, -1), (4, 4), (3, -2), (2, 2)$. An analyst wants to classify the dataset into two sets, one containing $\{(-4, 3), (-1, 0), (3, -2)\}$ and the other containing the rest. Can you find a separating hyperplane that will help the analyst? If yes, are there more than one such hyperplanes? Which hyperplane would you recommend using so that the hyperplane works even if the data points appear with an error ≤ 0.2 : for e.g. although $(4,0)$ is observed, the actual value of the data may be any (x_1, x_2) where $x_1 \in [3.8, 4.2]$ and $x_2 \in [-0.2, 0.2]$? What about error ≤ 0.5 ? (6)
3. Consider a divisible good in R . An envy-free allocation for 3 agents exists. True/False? Provide an argument for your answer and if yes, show the allocation procedure. Suppose the allocation is also expected

to maximize total valuation. Write down the optimization problem. Does it have a solution? Demonstrate with an example. (7)